

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2013-14

Paper: CSB-501 — Object Oriented Programming Using C++

Time: Three Hours

Maximum Marks: 70

Instructions: Answer any FIVE questions including Question 1 which is compulsory. The figures in the right-hand margin indicate full marks.

Question 1 (Compulsory).*[5 + 5 + 4 = 14]*

(a) State whether the following statements are TRUE or FALSE:

- (i) Polymorphism is extensively used in implementing inheritance.
- (ii) In C++, a class is a user-defined data type.
- (iii) The scope resolution operator can be used to uncover a hidden variable.
- (iv) The default visibility mode in C++ is private.
- (v) In C++, both virtual constructors and destructors are allowed.

(b) Fill in the blanks:

- (i) The operator `<<` is called the _____ operator.
- (ii) A _____ takes a reference to an object of the same class as itself as an argument.
- (iii) An _____ class is one that is not used to create objects.
- (iv) C++ uses a unique keyword called _____ to represent an object that invokes a member function.
- (v) C++ supports a mechanism known as _____ to achieve runtime polymorphism.

(c) Find errors, if any, in the following C++ statements:

- (i) `int *p = new;`
- (ii) `enum {green, yellow, red};`
- (iii) `for (i=1; int i<10; i++) cout << i << "\n";`
- (iv) `char name[3] = "USA";`

Question 2.*[5 + 5 + 4 = 14]*

(a) What are the unique advantages of an object-oriented programming paradigm?

(b) Write the output of the following C++ program:

```
#include <iostream>
using namespace std;
int a = 20;
```

```
int main() {
    int a = 40;
    {
        int k = a;
        int a = 50;
        cout << "INNER_BLOCK\n";
        cout << "k=" << k << "\n";
        cout << "a=" << a << "\n";
        cout << "::a=" << ::a << "\n";
    }
    cout << "OUTER_BLOCK\n";
    cout << "a=" << a << "\n";
    cout << "::a=" << ::a << "\n";
    return 0;
}
```

(c) What is a reference variable? What is the major use of this variable?

Question 3.

[6 + 3 + 5 = 14]

- (a) What do you mean by overloading of a function? When do we use this concept? Explain with an example.
- (b) When do we declare a member of a class static?
- (c) Write a program in C++ that shows how to use a common friend function to exchange the private values of two classes.

Question 4.

[5 + 5 + 4 = 14]

- (a) What is a constructor and a destructor? Also write the various types of constructors available in C++.
- (b) Write any five rules for overloading operators.
- (c) How many arguments are required in the definition of an overloaded unary operator? Explain with an example.

Question 5.

[5 + 5 + 4 = 14]

- (a) Explain the effect of inheritance on the visibility of class members using a diagram.
- (b) Write a program in C++ that demonstrates multilevel inheritance.
- (c) Write any four rules for virtual functions in C++.

Question 6.

[4 + 5 + 5 = 14]

- (a) Both `cin` and `getline()` functions can be used for reading a string. Comment.
- (b) What is a file mode? Describe the various file mode options available in C++.
- (c) Write a program in C++ that reads two files simultaneously.

Question 7. Write short notes on any two of the following:

[7 × 2 = 14]

- (a) Object-Oriented Analysis and Design (OOAD)
- (b) Class Templates
- (c) Exception Handling
- (d) Class Wizard